Synoptic CB: Macrophyte Fluxes

2024 Samples

2026-05-08

Contents

#Read in and collate all the files in the Processed Data folder

##Pull in Hydrago probe data

```
## All flux IDs are present in hydrago data
```

##pull TEROS data for the zone during the time of the fluxes and put it into another column so that we can use that too.

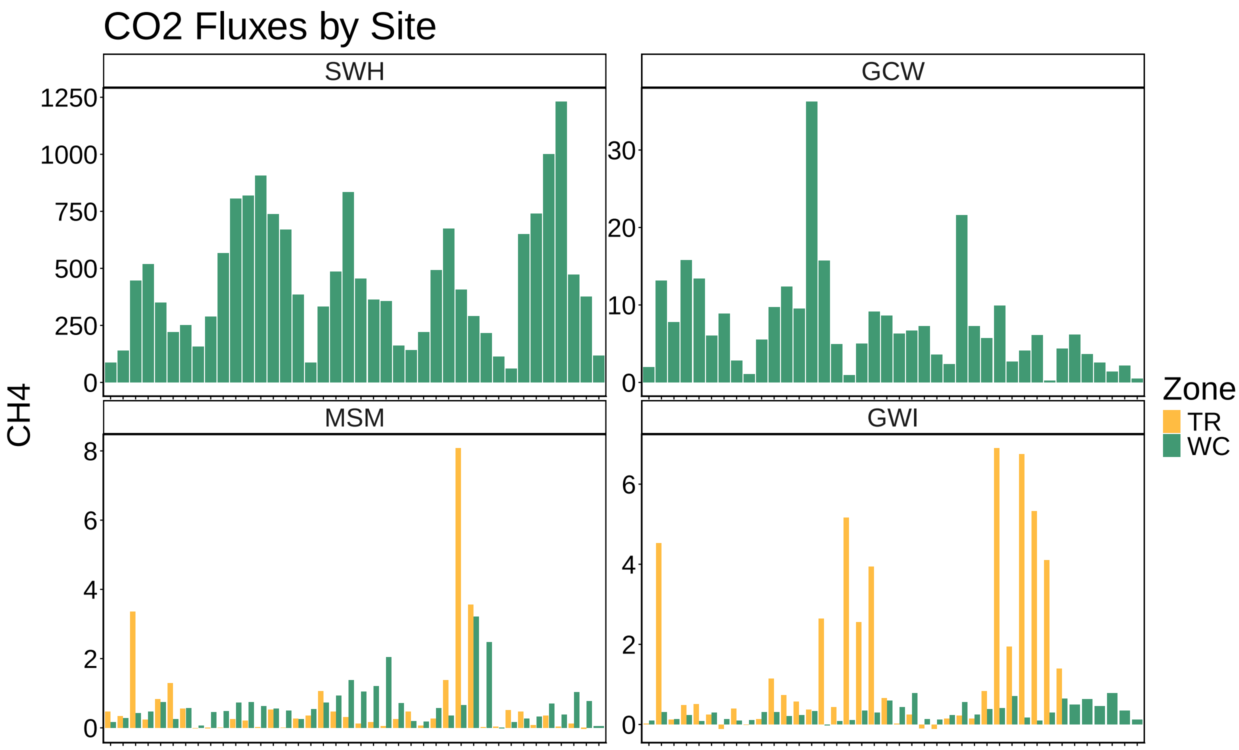
0.1 Match up the TEROS soil data and add it to the flux dataframe

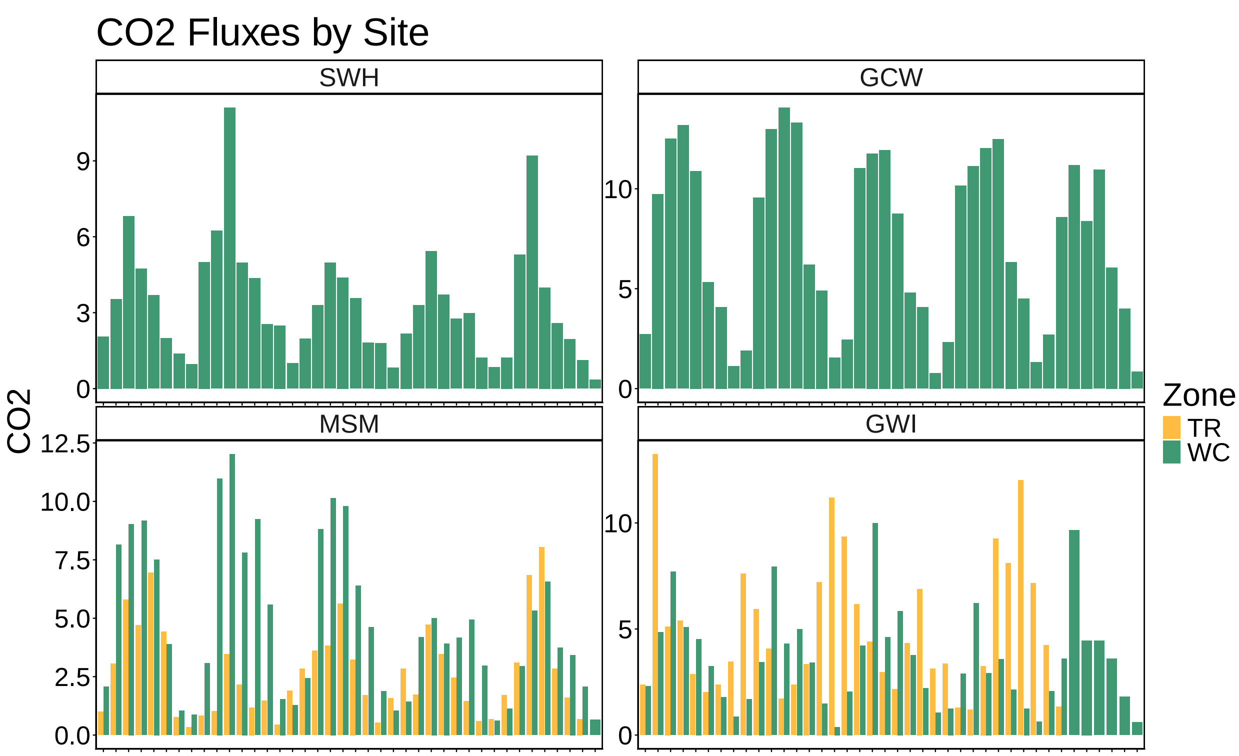
##Remove bad fluxes and calculate how many there were

```
## Number flux measurements removed: 2

## Number negative CO2 fluxes removed: 2

## Number messy fluxes removed: 1

## Percentage flux measurements removed: 0.85%

## Percentage negative CO2 fluxes removed: 0.85%

## Percentage messy fluxes removed: 0.43%
```

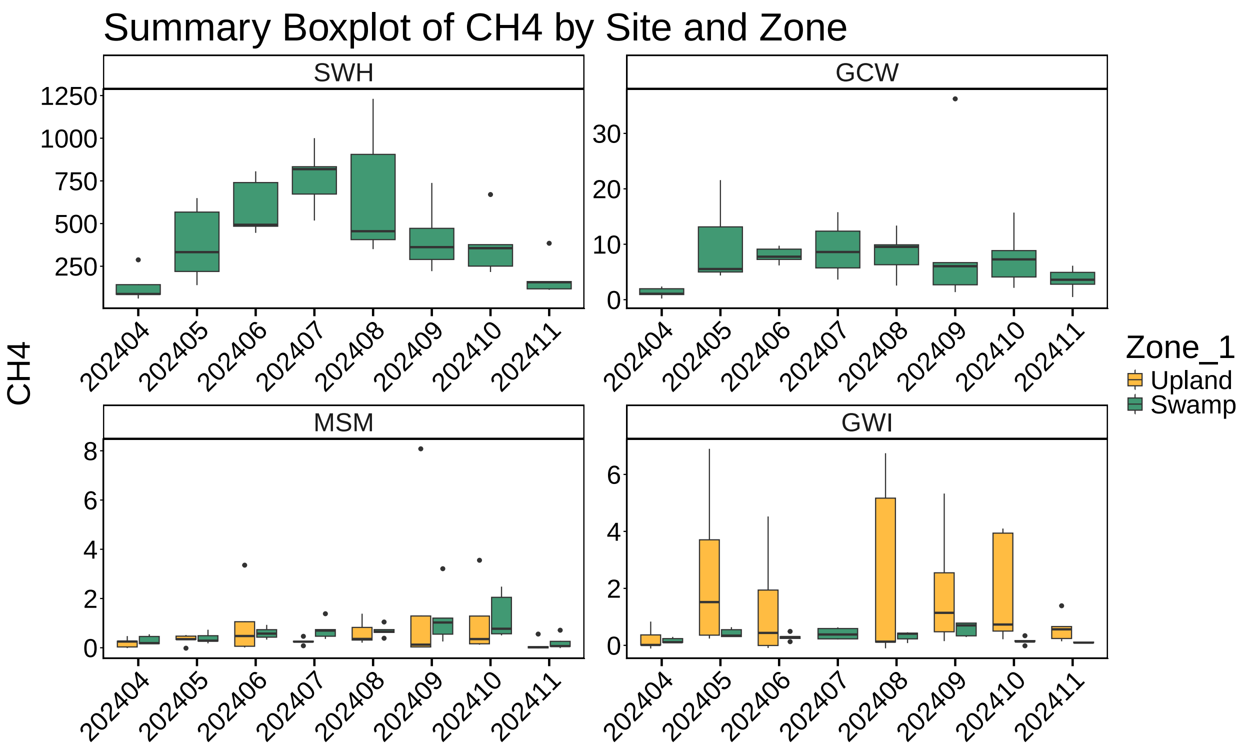
0.2 Write out files

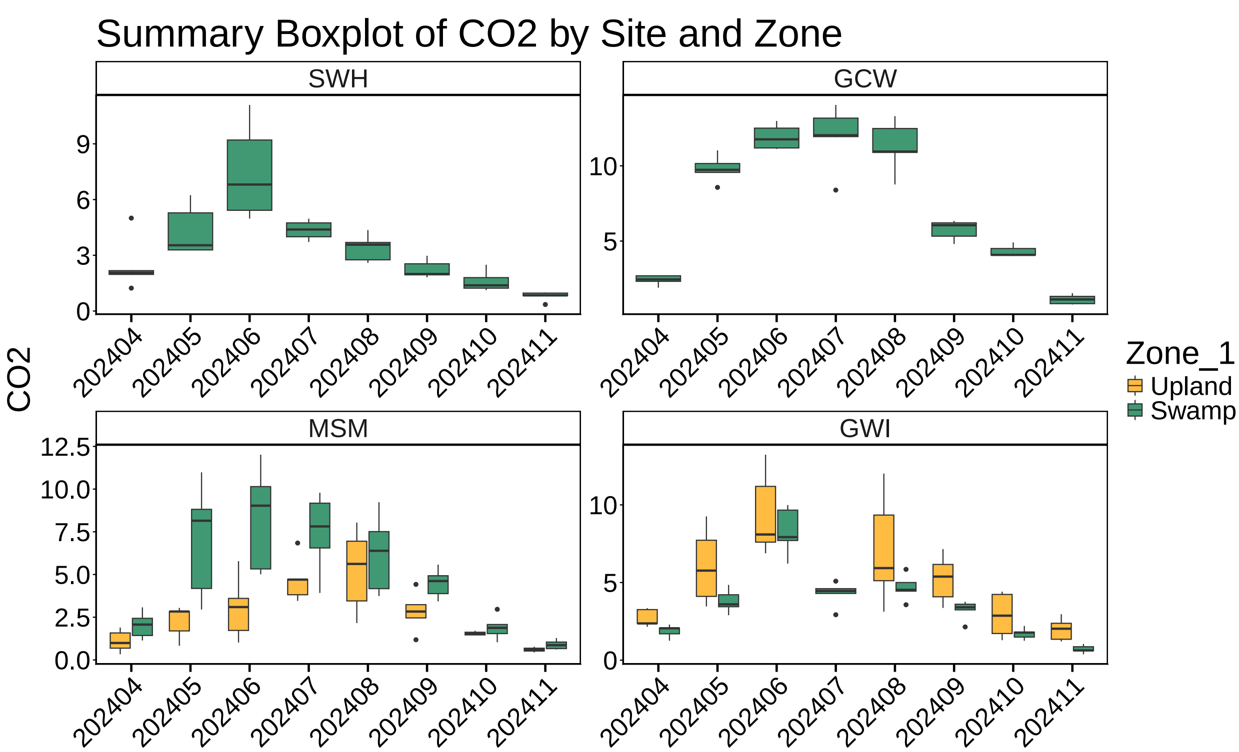
0.3 All Data by Year, Site and Zone



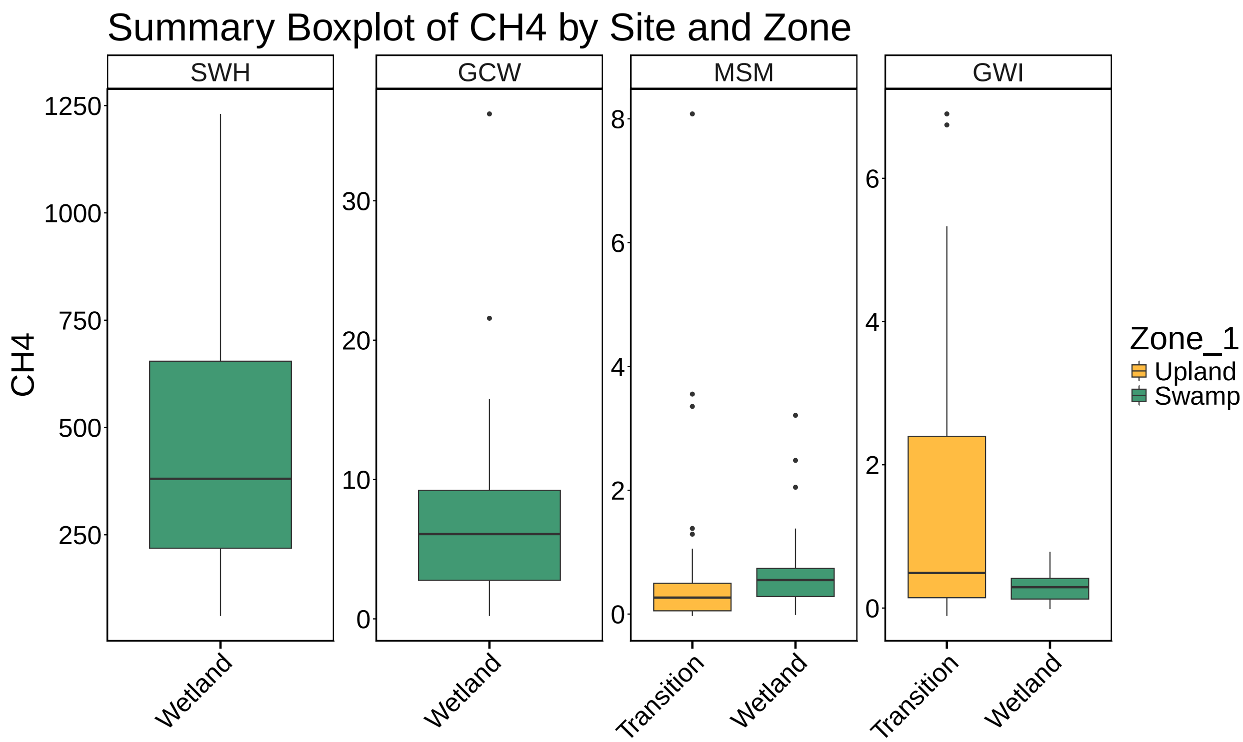


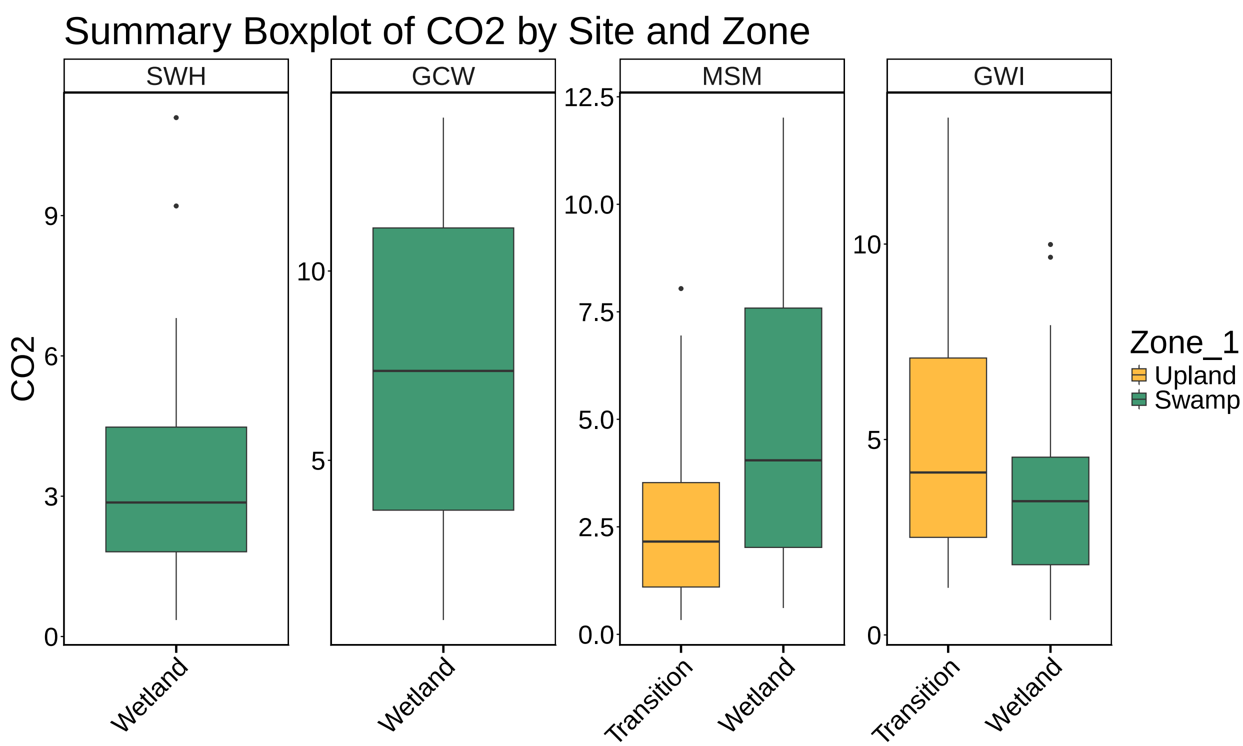
0.4 Boxplot summary by Date





0.5 Boxplot summary by Site





##Calculate summary metrics

0.6 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Wetland	60.8085475	1230.676594	440.3399282	0.3521706	11.094771	3.342001
GCW	Wetland	0.2071297	36.236250	7.3249032	0.7823220	14.050978	7.314527
MSM	Transition	-0.0310129	8.079763	0.6801219	0.3322192	8.040166	2.694096
MSM	Wetland	-0.0149613	3.210500	0.6819524	0.6116895	12.015151	4.786112
GWI	Transition	-0.1108124	6.898915	1.5376979	1.2049340	13.234715	5.024564
GWI	Wetland	-0.0154060	0.784701	0.3055206	0.3810429	9.988793	3.570166